

# Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation

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## Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to support code development and to polish the writing. All ideas and results are the author's own. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

## CRedit authorship contribution statement

**Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

## Ethics statement

The study uses only publicly released, anonymised data sets and involves no human subjects or personal data beyond them.

## Data availability

All data sets are public: Coat (<https://www.cs.cornell.edu/~schnabts/mnar/>), Yahoo! R3 (Yahoo! Webscope; the version distributed with the AutoDebias code, <https://github.com/DongHande/AutoDebias>), KuaiRand-Pure (<https://kuairand.com/>) and KuaiRec 2.0 (<https://zenodo.org/records/18164998>). Code, configurations and all result files are available at [https://github.com/vinhqdang/causal\\_learning\\_graph\\_recommendation](https://github.com/vinhqdang/causal_learning_graph_recommendation).